

Feature Importance & Model Comparison

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Machine Learning

Universe

- Pca: $\{\text{explained_variance_ratio}, \text{cumulative_variance}, \text{n_components_95pct}\}$
- Mutual Information: $\{\text{feature_scores}, \text{feature_ranking}, \text{selected_indices}\}$
- Rfe: $\{\text{feature_ranking}, \text{selected_indices}\}$
- Feature Overlap: $\{\text{mi_rfe_overlap}, \text{overlap_count}\}$
- Best Combination: $\{\text{model_name}, \text{feature_method}, \text{accuracy}, \text{accuracy_std}\}$

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

- **Data**: 800 samples, 20 features, binary target (~balanced classes)
- **True structure**: Only 5 features are informative, 3 are redundant, 12 are noise
- **Questions**: (1) How many components explain 95% of variance? (2) Which features matter most? (3) Which model + feature set combination is best?

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

- PCA explained variance**: How much total variance each principal component captures; cumulative
- Mutual information**: Information-theoretic measure of dependency between a feature and
- Recursive Feature Elimination (RFE)**: Iteratively removes features with the smallest impo
- Pipeline**: Standardized workflow of preprocessing, feature selection, and model fitting
- Cross-validation (5-fold)**: Splits data into 5 folds, trains on 4, tests on 1, rotates; provides
- Model comparison**: Evaluating multiple classifiers on identical data splits to identify the b

Approach

Suggested approach PCA + Mutual Information + RFE (Random Forest) + 4-Model 5-Fold CV

Complexity class:

Available tools: PCA + Mutual Information + RFE

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	PCA + Mutual Information + RFE (Random Forest) + 4-Model 5-Fold CV
Time:	5.6169s

Solution Details

Method: 1. PCA (Principal Component Analysis): Fit on all 20 features. Record explained variance per component and find the number of components needed for 95% cumulative variance. Reveals how much redundancy exists in the feature space.

2. Mutual Information feature selection: Score each feature's mutual information with the target. Rank all 20 features and select the top 8. Filter method -- fast, model-agnostic, detects nonlinear relationships.

3. Recursive Feature Elimination (RFE): Use a Random Forest estimator to recursively eliminate the least important features. Select the top 8. Wrapper method -- slower but accounts for feature interactions through the model.

4. Model comparison (5-fold CV): Compare Logistic Regression, Random Forest, SVM (RBF kernel), and k-NN on three feature sets: all 20 features, MI top 8, and RFE top 8. Each combination evaluated with 5-fold stratified cross-validation accuracy.

5. Best combination: Identify the model + feature set with the highest mean CV accuracy.

Verification

✓ Pca 95Pct Lt 20	Passed
✓ Informative Features Selected	Passed
✓ Rfe Features Selected	Passed
✓ Selected Beats Random	Passed
✓ Best Accuracy Gt 80Pct	Passed
✓ Cv Scores Stable	Passed

✓ All Passed

Passed

3. Interpretation

The Question

- **Data**: 800 samples, 20 features, binary target (~balanced classes)
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The Answer

A feasible solution was found.

What This Means

- PCA: Fewer than 20 components needed for 95% variance (dimensionality is compressible)
- Mutual information: Top 8 features recover at least 3 of the 5 true informative features
- RFE (Random Forest): Top 8 features also recover at least 3 of the 5 true informative features
- Best model accuracy: > 80% via 5-fold cross-validation
- Feature selection helps: Best selected-feature accuracy $\geq$ all-features accuracy - 0.05
- MI/RFE overlap: Substantial agreement between the two selection methods
- All 7 verification checks pass

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Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution